Deliberation Costs: LLM and VLM Negotiators for Head-On Robot Encounters in Single-Lane Corridors

Roshan George, Martin Glavin, Edward Jones, and Brian Deegan

***Abstract*—A companion study benchmarked classical negotiation policies for the head-on corridor encounter, where two autonomous mobile robots meet in a single lane and one must reverse into a refuge. Here we ask whether foundation-model agents can take that role, evaluating ten large language and vision-language models as negotiators under the same physics, prompts, and paired encounter seeds. In a centralized configuration, reasoning-capable models choose the right yielder in 92–94% of 200 encounters and reach -1.7 s mean excess delay *below* the best forced yielder, because they pick better refuges than the planner default. That capability is the deliberation itself: disabling chain-of-thought drops the same models to +11 to +30 s, below the hand-written rulebook. But when real inference latency is charged as decision time the ranking inverts — the best model's excess rises to +103 s and a third of some configurations deadlock — so classical cost bidding dominates any deployed comparison. For decentralized agents given egocentric views, disagreement is the dominant failure, and its mode is model-specific: one vision model defers on 84% of encounters while others race to proceed on $\sim$30%. Finally, injecting a fair, unspoofable random tiebreak eliminates every disagreement — and destroys deliberation: agents play the draw rather than negotiate, collapsing to coin-flip yielder choice. Keeping the same tiebreak invisible to the agents preserves their reasoning; stated mechanisms shape agent behavior, not just outcomes. Code and per-episode results are released.**

***Index Terms*—Large language models, vision-language models, multi-robot systems, negotiation, mechanism design, autonomous mobile robots.**

I. Introduction

WHEN two autonomous mobile robots meet head-on in a corridor too narrow to pass, someone must reverse into a pull-in bay or out of the lane entirely. A companion benchmark [1] compared reactive, fixed-priority, rule-based, reservation, cost-bidding, group-bidding, Clarke-pivot, and learned negotiators under identical physics and communication models, and found that cost bidding with accurate costs tracks the best forced yielder — a hindsight reference that tries both yield decisions — while degrading gracefully under bid noise and misreporting. That benchmark deliberately excluded one agent family: foundation models that read a textual or visual description of the encounter and decide in natural language.

Language-model negotiators are attractive for a specific reason: the corridor decision is a small, well-posed optimization — pick the yielder minimizing total delay — yet it is exactly the kind of social, two-party choice that language models are claimed to handle. Warehouses are increasingly populated by robots from multiple vendors with no shared planner; an agent that can reason over a broadcast state and pick a sensible yielder without bespoke protocol engineering would be valuable. The question is whether current models can, and what it costs.

We attach a large-language-model (LLM) and vision-language-model (VLM) policy adapter to the companion benchmark and evaluate ten models — local 9–49B-parameter weights on a single Grace-Blackwell node plus a hosted 120B endpoint — across controlled arms that isolate decision quality, inference cost, input modality, and coordination structure. The same encounter seeds serve every arm, so comparisons are paired. Our contributions:

1) a prompt-and-parse LLM/VLM negotiator for the corridor encounter, with centralized (*mediator*) and decentralized (*agents*) modes, text and image inputs, JSON-schema constraint, deterministic caching, and latency charged into the simulator as decision time;
2) a model screen showing reasoning-capable models *exceed* the forced-yielder reference through better refuge selection — and that this capability vanishes when chain-of-thought is disabled;
3) a deployed-cost analysis showing measured inference latency reverses the ranking: deliberating agents pick the right yielder but freeze long enough to lose the benefit, and slower configurations deadlock;
4) an egocentric decentralized study in which disagreement — not decision quality — is the dominant failure, with model-specific failure personalities (deference vs. self-serving attribution);
5) a protocol-default experiment: a fair, unspoofable random tiebreak eliminates all disagreement while simultaneously destroying the incentive to deliberate — agents play the draw instead of the encounter. Keeping the identical tiebreak invisible preserves reasoning entirely.

II. Related Work

Language models have been applied to negotiation [2], [3], multi-agent coordination games, and robot task planning; [4] uses a foundation model for high-level deadlock resolution. These studies evaluate agents in bespoke environments with bespoke metrics; none compares foundation-model negotiators against a classical optimal baseline under identical

All authors are with the Connaught Automotive Research (CAR) group and the Ryan Institute, University of Galway, Ireland (e-mail: brian.deegan@universityofgalway.ie).

DRAFT, not for distribution.

physics, which the companion benchmark [1] provides. Vision-language models are increasingly used for robot perception and instruction following, but we are not aware of a controlled comparison of text, image, and egocentric-image inputs for an inter-robot *decision* task.

Our tiebreak result connects to game theory: public randomization resolves anti-coordination equilibria in correlated-equilibrium terms [5], and commitment-based coin-flipping without a trusted dealer is a classical cryptographic result [6]. In bargaining theory, the disagreement point shapes negotiated outcomes [7]; we find its language-model analogue is stronger still — a common-knowledge default does not merely bound the negotiation, it *becomes* it.

III. Benchmark and Agents

A. The corridor encounter

We use the companion benchmark [1] unchanged. Two TurtleBot3-Waffle-class unicycle robots (0.26 m/s forward, 0.15 m/s reverse, radius 0.15 m) approach head-on in an 8 m, 0.65 m-wide single-lane corridor with a pull-in bay and end-room refuges. On conflict detection each robot broadcasts a frozen *snapshot*: position, heading, speed, goal, corridor entry time, urgency, and payload state. A policy receives the two snapshots and returns a *yielder*, a *proceeder*, and a refuge; the yielder reverses into or toward the refuge, waits, then continues. The 2D unicycle simulator runs Monte Carlo encounters; we use the single-bay and two-bays layouts, 100 encounters per layout per condition (200 total), drawn from held-out seeds. Metrics are success, collision, deadlock, urgency-weighted excess delay against the best forced yielder (a reference that replays both yield assignments and takes the better), and *right-yielder rate* — the fraction of encounters matching the reference's yielder, excluding ± 0.5 s ties. A rulebook decides every encounter that the LLM does not resolve, so all runs succeed; the fallback fraction reports how often the model was unreachable, unparseable, infeasible, or in disagreement.

B. LLM policy adapter

The `llm` policy issues OpenAI-compatible chat requests at decision time. In *mediator* mode one call describes both snapshots and asks for the yielder; in *agents* mode each robot is prompted separately — "you are robot X" — and answers yield or proceed, with agreement required and disagreement falling back. Two information levels exist: *raw* gives the natural-language state (positions, headings, speeds, urgency, refuges with distances), while *plans* additionally includes each robot's predicted yield delay per refuge from the kinematic planner — the same numbers cost bidding uses. Responses are JSON-schema constrained when the server supports it; any error, parse failure, infeasible refuge, or disagreement falls back to the rulebook with a classified reason. Replies are cached deterministically on the hashed request, so runs are reproducible and resumable. Two cost modes isolate quality from deployment: `think=0` charges no decision time (pure decision quality), while `think=measured` freezes the robot for the real response latency, or the slower of the two agent calls — the quantity a deployed negotiator would actually pay.

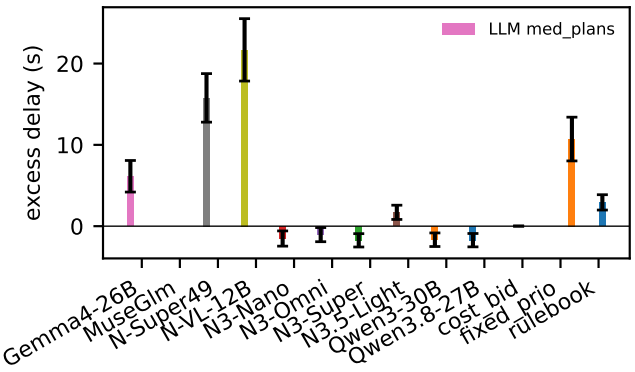


Fig. 1. Model screen, mediator+plans arm, think=0 (200 paired encounters). Negative excess beats the best-forced-yielder reference.

C. Vision arms

Two renderers feed VLMs the encounter as an image. *Global*: a top-down map showing walls, refuge parking marks, both robots as labeled discs with heading arrows, urgency/load annotations, and goal stars. *Egocentric*: the same render clipped to each robot's forward view (-1.5 to $+6.5$ m along its heading) — the agents mode's onboard-camera analogue, so each robot sees only what it would sense, while the broadcast text still carries the peer's state. Modality arms compare text-only, image-only, and text+image on the global view, and text vs. egocentric image on the decentralized mode.

D. Models

We screen ten deployments (Table I): eight models served locally on one NVIDIA GB10 node (vLLM, NVFP4 quantization, per-family reasoning parsers) plus hosted `nemotron-3-super-120b`. Reasoning models emit a chain-of-thought before answering; a per-request switch (`enable_thinking`) disables it where supported — we verified on-cache that the reasoning field is empty and latency drops 10–30$\times$ when off. *Thinking-off* means the model emits the decision directly; *thinking-on* means it writes a trace first.

E. Protocol

Every arm runs the same 200 encounters per layout pair and the same seed draw, so all model comparisons are paired. Screen arms use mediator+plans, `think=0`; the matrix expands mode, info, thinking, and think-cost on the top three. Decentralized arms use agents mode; vision arms add the renderers. Fallback composition, per-layout splits, and full per-episode answers (including reasoning traces) are in the released data.

IV. Results

A. Model screen

Figure 1 screens all models on the mediator+plans arm (`think=0`). Reasoning mixture-of-experts models cluster tightly: qwen3.8-27B, qwen3-30B, and hosted nemotron-3-super reach -1.7 s mean excess, nemotron-nano -1.5 s, and

TABLE I
MODELS SCREENED (LOCAL VLLM ON ONE GB10 NODE + HOSTED ENDPOINT).

Model	Params	Notes
Nemotron-3.5-Lightning	30B-A3B	reasoning, MTP spec.
Nemotron-3-Nano	30B-A3B	reasoning
Nemotron-3-Nano-Omni	30B-A3B	reasoning, VL
Llama-3.3-Nemotron-Super	49B	dense
Nemotron-Nano-VL	12B	VL
Qwen3-30B-A3B	30B-A3B	reasoning
Qwen3.8-27B	27B	reasoning, VL
Gemma4-26B-A4B	26B-A4B	VL
Muse-Glimmer	30B	experimental
nemotron-3-super-120b (hosted)	120B-A12B	NVIDIA API

nemotron-omni -1.0 s — all *below* the forced-yielder reference (0.0 s), because the reference picks the yielder but takes the planner's default refuge while the LLMs sometimes pick a better one. Right-yielder rates are 92–93%. Nemotron-lightning underperforms (1.7 s, 80%) with 66% parse fallback — its traces frequently exhaust the 4096-token cap. The non-reasoning tier is starkly worse: gemma4-26B at 6.1 s (75%), dense nemotron-super-49B at 15.8 s (42%), and 12B nemotron-vl at 21.7 s (38%) — worse than the hand-written rulebook (2.9 s, 72%) and approaching fixed priority (10.7 s). A 1.5B smoke-test model collapsed entirely (25–47 s, 25–30% right). Deliberating ∼30B models thus meet or beat the benchmark's own reference; non-deliberating models do not negotiate, they guess.

B. Deliberation is the capability

To test whether the reasoning trace carries the capability, we rerun the top models with `enable_thinking=false` (verified empty reasoning fields, 9.3 s to 4.2 s mean call latency on qwen3.8). On identical encounters, qwen3.8-27B drops from -1.7 s (93% right) to $+11.1$ s (66%) on plans and $+23.2$ s (40%) on raw; qwen3-30B collapses further, to $+22.5$ s (35%) and $+29.9$ s (36%) — below fixed priority and approaching the 1.5B failure mode (Fig. 2). The weights do not encode the corridor decision; the chain-of-thought computes it. Notably, even thinking-off agents receive identical plans numbers in the prompt — they fail not for lack of the cost estimate but for lack of the arithmetic.

C. The deployed cost

Decision quality is only half the story: a negotiator that takes 30–130 s to answer stalls the robot mid-corridor. Charging measured latency as decision time inverts the ranking (Fig. 3). qwen3-30B mediator+plans — -1.3 s in the quality arm — rises to $+103.1$ s excess with a 31.1 s mean think, while *right-yielder rate stays at 92%*: the decisions are still correct, the robot simply pays for the deliberation in dead time. Worse, latency compounds into failure: on the raw arm, thinking-on+measured succeeds in only 64% of encounters (88% parse-fallback traces exceeding the token cap under pressure), leaving a third deadlocked. Thinking-off restores speed but not quality: a serial rerun gives 0.7 s calls yet 23.6 s excess — latency is no longer the binding constraint, deliberation loss

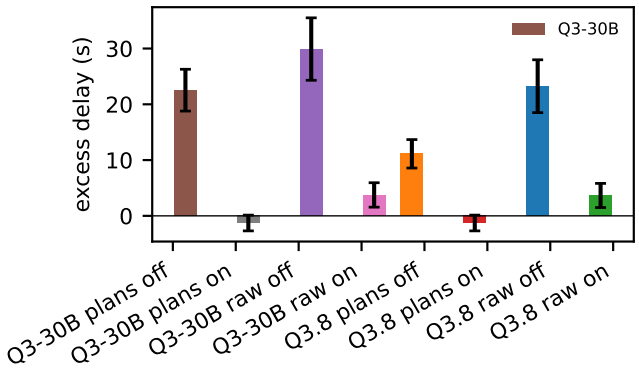


Fig. 2. Reasoning on vs off, mediator arms (paired seeds; on cells n=50). Disabling chain-of-thought collapses decision quality.

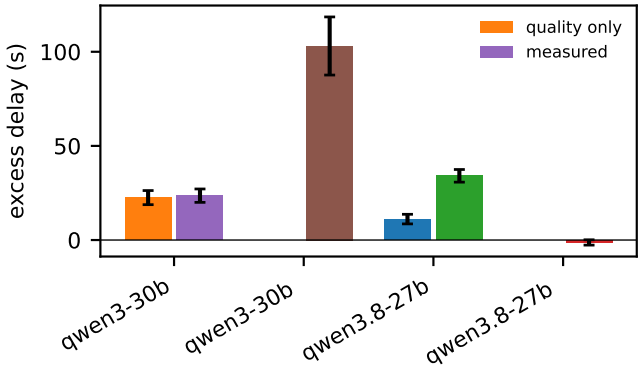


Fig. 3. Quality-only vs measured-latency excess, mediator+plans. Charging real inference time inverts the ranking; right-yielder is preserved.

is. The deployed verdict is unambiguous: at current speeds, cost bidding (sub-millisecond, 0.0 s) wins any real-time comparison; an LLM negotiator is viable only if deliberation is off the critical path — precomputed, asynchronous, or sub-second.

D. Vision input

Four VL-capable models ran text-only, image-only (global render), and text+image mediator arms, thinking-off (Fig. 4). Image-only beats text-only on *every* model — gemma4 3.9 vs 6.6 s, qwen3.8 9.0 vs 14.9 s, nemotron-omni 14.2 vs 16.4 s, vl-12b 17.0 vs 22.1 s — so visual grounding carries corridor geometry better than a numeric description for these models. Yet text+image is worse than image alone on three of four (e.g. vl-12b 30.0 s, omni 20.6 s): the modalities interfere rather than complement. All image arms still sit far below reasoning-on text (-1 to -1.7 s) — vision substitutes for deliberation only partially.

E. Egocentric decentralized agents

Chance encounters between unaffiliated robots motivate the agents mode: each robot gets only its egocentric view plus the peer's broadcast and decides independently. Here *disagreement*, not decision quality, dominates — and its

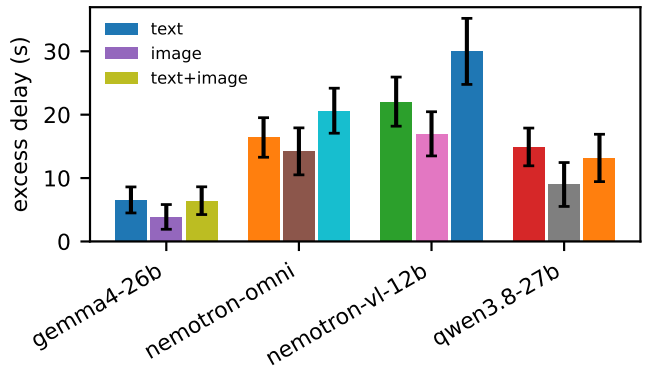


Fig. 4. Modality arms, mediator, thinking-off. Image-only beats text-only on all four VL models; combining modalities interferes.

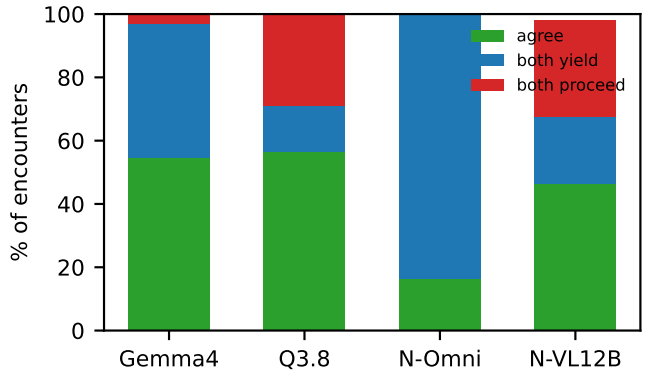


Fig. 5. Action-agreement modes, egocentric agents arm. Disagreement mode is a model-specific personality: deference (both-yield) vs. self-serving conflict (both-proceed).

mode is a model-specific personality (Fig. 5). nemotron-omni defers pathologically: both robots volunteer to yield on 84% of encounters (deadlock-by-politeness, rulebook-rescued). qwen3.8-27B and vl-12b instead race ahead: both-proceed conflicts on 29% and 31% — the reasons show self-serving attribution, each robot citing the *same* proximity facts to award the yield to the *other*. gemma4 is closest to healthy: 55% agreement, mostly both-yield errors. On agreed encounters the right-yielder rate is only 58–71%, and *wrong-yielder agreements are the most expensive failure observed* (13.7–38.0 s) — correlated wrongness executes without any fallback. Ego images modestly beat raw text for weaker agents (vl-12b 5.3 vs 7.4 s; qwen3.8 8.9 vs 10.0 s) but cannot supply the coordination convention that is missing.

F. The protocol default

Disagreement calls for a tiebreak — but *how* it is presented matters. We inject a fair, unspoofable draw: each robot's "card" is a hash bound to its id and the encounter time, computed identically from broadcast data, so neither can spoof it; the larger card wins on conflict. Three arms isolate presentation from resolution (Fig. 6).

Visible draw (rule in the prompt): disagreements vanish entirely — and so does deliberation. Agents preempt the draw

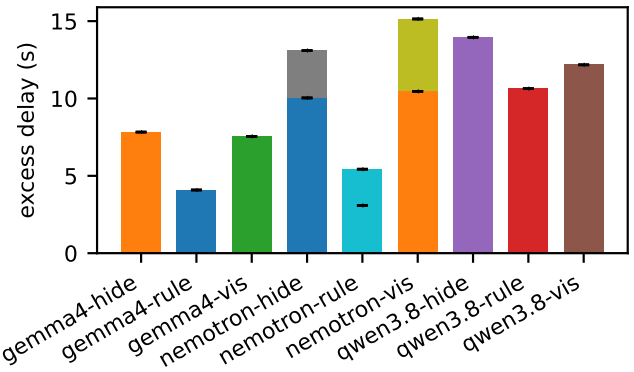


Fig. 6. Protocol-default triad: rulebook fallback vs hidden vs visible random tiebreak (egocentric agents, paired seeds).

rather than negotiate: right-yielder rate collapses to 48–58% (coin-flip), and omni's trace shows it plainly — "A's card is smaller than B's, so if both yield or proceed, B would proceed." The common-knowledge default *becomes* the policy.

Hidden draw (identical prompts, draw resolves silently): answer distributions are *identical* to the rulebook arm (e.g. omni 170/27/3 action-count split in both) — deliberation is fully preserved. Conflicts now resolve by chance rather than optimally: 45–51% right on tiebroken encounters, 11–14 s excess, versus the rulebook's 99% and 2–5 s.

The triad separates incentive from mechanism: agent behavior tracks what they *believe* the default is, not what it is — and a fair default is only safe when it is worse than what agents can negotiate, or invisible.

V. Discussion

The picture that emerges is more optimistic than the latency numbers suggest — and more cautionary than the quality numbers. For a fleet setting where a shared decision service is acceptable, a reasoning mediator with plans information is already at or beyond the benchmark's forced-yielder reference; the engineering work is making deliberation asynchronous (predictive negotiation, speculative caching of likely encounters, or a fast tier for routine cases). For true chance encounters between unaffiliated robots, decentralized foundation-model agents are not yet deployable: disagreement is structural, and a stated tiebreak fixes it only by destroying the reasoning that makes the agents useful. The practical guidance for protocol design is narrower than expected: keep defaults invisible, or make the negotiated outcome strictly better than the default — a visible fallback is a policy, not a safety net.

A. Limitations

All results are simulation; the Gazebo transfer validated in [1] covers classical policies, not the LLM arms. Two layouts and one encounter type bound the generality. The hosted 120B endpoint was unreliable under study load (∼100% errors on several cells), so its screen number is the only trusted hosted datapoint. Thinking-off is honored on the models tested but is not universal across serving stacks, and several models'

traces exceed the 4096-token cap on the raw prompt (the parse-fallback counts reflect this). Finally, agents answer a single-shot prompt; multi-turn dialogue, tool use (querying the planner), and selfish-utility prompts are untested arms.

VI. Conclusion

Reasoning-capable language models *can* negotiate head-on robot encounters at or beyond a strong classical reference — the capability is real, and it is the deliberation, not the parameters. But deployed viability is governed by latency, not decision quality: at current inference speeds a millisecond bid mechanism wins by an order of magnitude. For decentralized agents, coordination conventions — not perception — are the bottleneck: models disagree in personality-specific ways that better sensing does not fix. And the mechanism-design lesson generalizes: for LLM agents, the stated default is not a floor but a magnet. Design protocols accordingly.

References

[1] Anonymous, "Who backs into the bay? benchmarking negotiation policies for head-on robot encounters in single-lane corridors," 2026, companion submission.

[2] T. Godfrey, W. Hunt, and M. D. Soorati, "MARLIN: Multi-Agent Reinforcement Learning Guided by Language-Based Inter-Robot Negotiation," arXiv preprint, 2024. [Online]. Available: https://arxiv.org/abs/2410.14383

[3] V. Mahadevan, S. Zhang, and R. Chandra, "GameChat: Multi-LLM Dialogue for Safe, Agile, and Socially Optimal Multi-Agent Navigation in Constrained Environments," in *2025 IEEE International Symposium on Multi-Robot and Multi-Agent Systems (MRS)*, 2025, pp. 1–7. [Online]. Available: https://doi.org/10.1109/mrs66243.2025.11357267

[4] K. Garg, S. Zhang, J. Arkin, and C. Fan, "Foundation Models to the Rescue: Deadlock Resolution in Connected Multi-Robot Systems," arXiv preprint, 2024. [Online]. Available: https://arxiv.org/abs/2404.06413

[5] R. J. Aumann, "Subjectivity and correlation in randomized strategies," *Journal of Mathematical Economics*, vol. 1, no. 1, pp. 67–96, 1974.

[6] M. Blum, "Coin flipping by telephone: A protocol for solving impossible problems," *SIGACT News*, vol. 15, no. 1, pp. 23–27, 1983.

[7] J. F. Nash, "The Bargaining Problem," *Econometrica*, vol. 18, no. 2, p. 155, 1950. [Online]. Available: https://doi.org/10.2307/1907266